

Fig. 3: MIMD architecture

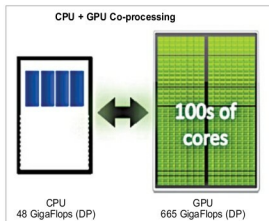


Fig. 4: Heterogeneous computing with GPUs

program multiple-data (SPMD) and multiple-program multiple-data (MPMD). Single-chip cell broadband engine architecture consists of a traditional CPU core and eight SIMD accelerator cores. It is a very flexible architecture, where each core can run separate programs in MPMD fashion and communicate through a fast on-chip bus. Its main design criteria has been to maximise performance whilst consuming minimal power. It defines a new processor structure based upon the 64-bit Power Architecture technology, but with unique features directed toward distributed processing and media-rich applications.

The cell broadband engine architecture defines a single-chip multi-processor consisting of one or more power processor elements (PPEs) and multiple high-performance SIMD synergistic processor elements

(SPEs). For example, a GPU with 30 highly multi-threaded SIMD accelerator cores in combination with a standard multicore CPU.

The GPU has a vastly superior bandwidth and computational performance, and is optimised for rerunning SPMD programs with little or no synchronisation. It is designed for high-performance graphics, where data throughput is the key. FPGA can also incorporate regular CPU cores on-chip, making it a heterogeneous chip by itself. FPGAs can be viewed as user-defined ASICs that are reconfigurable. These offer fully deterministic performance and are designed for high throughput, for example, in telecommunication applications.

Accelerator cores are designed to maximise performance, given a fixed power or transistor budget. These use fewer transistors and run at lower frequencies than traditional CPUs.

Algorithms such as finite-state machines and other intrinsically serial algorithms are most suitable for single-core CPUs running at high frequencies. Parallel algorithms such as Monte Carlo simulations, on the other hand, benefit greatly from many accelerator cores running at a lower frequency. Most applications consist of a mixture of such serial and parallel tasks, and ultimately perform best on heterogeneous architectures.

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